Once $\boldsymbol{\Lambda}$ is known, then this gives $\boldsymbol{\lambda}_n$ which in turn gives, after using (4.15), $\boldsymbol{p}_{n+1}$ and finally (4.14) yields $\boldsymbol{q}_{n+1}$.

Since $\boldsymbol{Q}_n$ and $\boldsymbol{G}_n$ are known at step n, Eq. (4.17) is an equation for the m components of $\boldsymbol{\Lambda}$. This may be solved by various iterative strategies. The simplest is Gauss-Newton iteration, which may be written

$$\boldsymbol{\Lambda}^{(l+1)} = \boldsymbol{\Lambda}^{(l)} + \left[\boldsymbol{g}'(Q^{(l)})\boldsymbol{M}^{-1}\boldsymbol{G}_n^T\right]^{-1} \boldsymbol{g}(\boldsymbol{Q}^{(l)}), \tag{4.18}$$

$$\boldsymbol{Q}^{(l)} := \boldsymbol{Q}_n - \boldsymbol{M}^{-1}\boldsymbol{G}_n^T\boldsymbol{\Lambda}^{(l)}, \tag{4.19}$$

which might be initialized with $\boldsymbol{\Lambda}^{(0)} = \boldsymbol{0}$. A more sensible way to write this is

$$\boldsymbol{R}^{(l)}\Delta\boldsymbol{\Lambda}^{(l)} = \boldsymbol{g}(\boldsymbol{Q}^{(l)}), \qquad \boldsymbol{R}^{(l)} \stackrel{\text{def}}{=} \boldsymbol{g}'(Q^{(l)})\boldsymbol{M}^{-1}\boldsymbol{G}_n^T,$$

with $\Delta\boldsymbol{\Lambda}^{(l)} = \boldsymbol{\Lambda}^{(l+1)} - \boldsymbol{\Lambda}^{(l)}$. The usual way of solving such equations is by Gaussian elimination, which will require re-evaluation of $\boldsymbol{R}$ and refactorization at each iteration. However, assuming that the steps are sufficiently small and that $\boldsymbol{g}'$ is sufficiently smooth, we may be able to fix $\boldsymbol{R}$ at the beginning of the timestep, factor it once only, and then solve at each iteration with the same factors. Moreover, in case we have chosen $\Lambda^{(0)} = 0$, we have

$$\boldsymbol{R}^{(0)} = \boldsymbol{G}_n\boldsymbol{M}^{-1}\boldsymbol{G}_n^T,$$

which is a symmetric matrix, and it is well known that in such a case, where a Cholesky factorization is available, both efficiency and reliability of the elimination process are favorably improved.

When the solution is unique, the constraints sufficiently smooth, and the determinant of $\boldsymbol{R}$ is bounded well away from zero, Newton's method has a very rapid asymptotic convergence property. It is possible to show that, if $\boldsymbol{\Lambda}_*$ is the solution of (4.17), then sufficiently near the zero we have

$$\|\boldsymbol{\Lambda}^{(l+1)} - \boldsymbol{\Lambda}_*\| \leq C\|\boldsymbol{\Lambda}^{(l)} - \boldsymbol{\Lambda}_*\|^2.$$

Thus, if C is of moderate size, then two digits of accuracy becomes four, four becomes eight, etc. This rapid rate of convergence is only seen sufficiently near the root and only if $\boldsymbol{R}$ is re-evaluated at each stage of iteration. When $\boldsymbol{R}$ is evaluated only once per step, the asymptotic convergence of Newton's method is linear, i.e. we have instead

$$\|\boldsymbol{\Lambda}^{(l+1)} - \boldsymbol{\Lambda}_*\| \leq \hat{C}\|\boldsymbol{\Lambda}^{(l)} - \boldsymbol{\Lambda}_*\|,$$

sufficiently near to the solution. To stop the Newton iteration, we may check the stopping condition

$$\|\boldsymbol{\Lambda}^{(l+1)} - \boldsymbol{\Lambda}^{(l)}\| < \text{tol},$$

or else check the constraint residuals, or do both.